



Bangladesh Army University of Engineering & Technology (BAUET)

Department of Civil Engineering

B.Sc. Engg. 1st Year 2nd Semester, Mid-Term Examination, Winter-2025

Course Code: MATH 1201

Full Marks: 20

Course Title: Differential Equations and Statistics

Time: 01 Hour

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer two questions, including Q.1.
3. Assume reasonable data if necessary.

R. H. H.

Marks

Q.1 (a) Define each term with an example "order and degree of a differential equation". 2
[CLO1, PLO1, C1]

(b) Identify the following equations whether they are ordinary or partial differential equation, then indicating their order and degree, also identify whether they are linear or non-linear: 3×2
[CLO1, PLO1, C2]

(i) $\frac{\partial^6 v}{\partial x^4 \partial t^2} + \frac{\partial^5 v}{\partial t^3 \partial x^2} + \frac{\partial^4 v}{\partial t^4} = 0$, (ii) $\sqrt{\frac{dx}{dy}} = \sqrt{\frac{d^2 x}{dy^2} - x}$, (iii) $\left(\frac{d^4 y}{dx^4}\right)^3 + \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = e^x$.

(c) Obtain a differential equation satisfied by family of circles $x^2 + y^2 = a^2$, a being an arbitrary constant. 2
[CLO2, PLO2, C3]

Q.2 Solve the following first order and first-degree ordinary differential equations: 3+4+3
[CLO2, PLO1, C3]

(i) $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$

(ii) $\frac{dy}{dx} + \frac{y-2x}{4y+3x} = 0$

(iii) $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$

Q.3 (a) Write down the general form of first order linear ordinary differential equation. Obtain the solution of the differential equation $\sin x \frac{dy}{dx} + 2y = \cos x$. 1+5
[CLO2, PLO1, C3]

(b) Determine the solution of ordinary differential equation $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$. 4
[CLO2, PLO1, C3]

21 + 2(14 + 3)
-> 19 ID
(14 + 3) dy + (1 - 2x) dx = 0

1/2 sec^2 y/2
1/2 tan^2 y/2

*1/2 = 10 * e^x * e^y - e^x * e^y*
*= e^x * (e^y - e^y)*
= e^x (e^y - e^y)

Bangladesh Army University of Engineering & Technology (BAUET)

Department of Civil Engineering (CE)

B.Sc. Engineering 1st Year 2nd Semester, Midterm Examination/Assessment, Winter-2025

Course Code: CHEM-1201

Full Marks: 20

Course Title: Chemistry-II

Time: 1:00 Hour

N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer Two questions, including Q. 1
3. Assume reasonable data if necessary.

		Marks
Q.1	(a) Explain the terms pig iron and cast iron. [CO1, PO1, C1, K1]	2
	(b) Discuss the Bessemer process for steel production. [CO2, PO2, C2, K1]	4
	(c) Demonstrate the chemical reactions that occurred at different heights in the blast furnace during the production of pig iron. [CO3, PO3, C3, K3]	4
Q.2	(a) Define the terms sound cement and unsound cement. [CO1, PO1, C1, K1]	2
	(b) Name the raw materials of cement. [CO2, PO2, C1, K1]	4
	(c) With a neat sketch, explain the construction and working process of the rotary kiln in cement manufacturing. [CO3, PO3, C3, K3]	4
Q.3	(a) Define the term clinker of cement. Describe its role in cement manufacturing. [CO1, PO1, C1, K1]	2
	(b) Name the most important minerals of cement. [CO1, PO1, C1, K1]	4
	(c) Discuss the standard testing parameters used to evaluate the quality of cement. [CO2, PO2, C2, K1]	4

Bangladesh Army University of Engineering & Technology (BAUET)

Department of Civil Engineering (CE)

B.Sc. Engg. 1st Year 2nd Semester, Mid-Term Examination, Winter-2025



Course Code: EEE- 1211

Full Marks: 20

Course Title: Basic Electrical Technology

Time: 01 Hour

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer **Two** questions **including Q.1.**
3. Assume reasonable data if necessary.

Q.1 (a) Describe Kirchhoff's Voltage Law (KVL) with appropriate circuit examples and neat sketch. [CO1, PO1] 3

(b) In the following circuit shown in Fig. 01(b), a 100V DC source is connected to the ab terminal of the circuit. [CO1, PO1] 7

- i. Determine the equivalent resistance with respect to terminals a-b.
- ii. Determine the current supplied by the source.

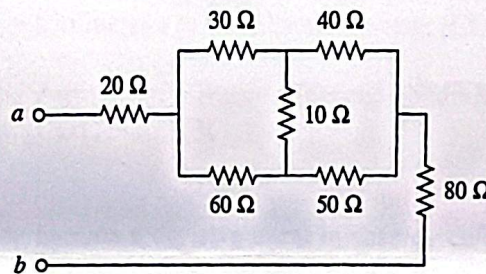


Fig. 01(b)

Q.2 (a) State Ohm's Law and explain it mathematically. [CO1, PO1] 2

(b) Identify the number of dependent and independent sources in the circuit, and then analyze the circuit shown in Fig. 02(b) to determine the branch currents I_1 , I_2 , and I_3 using mesh analysis. [CO2, PO2] 8

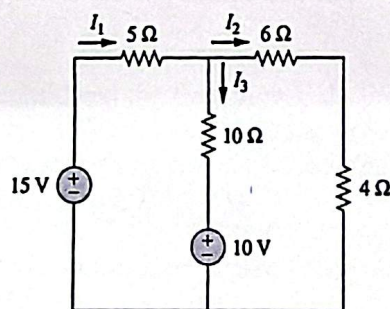


Fig. 02(b)

Q.3 (a) Explain the differences between passive and active circuit elements. [CO1, PO1] 2

$$10V - 10I_a$$
$$10V - 10I_a$$
$$10V = 10I_a$$

- (b) Identify the number of active and passive elements present in the circuit shown in Fig. 03(b). Analyze the circuit using nodal analysis to obtain the current flowing through the $4\ \Omega$ resistor. [CO2, PO2]

8

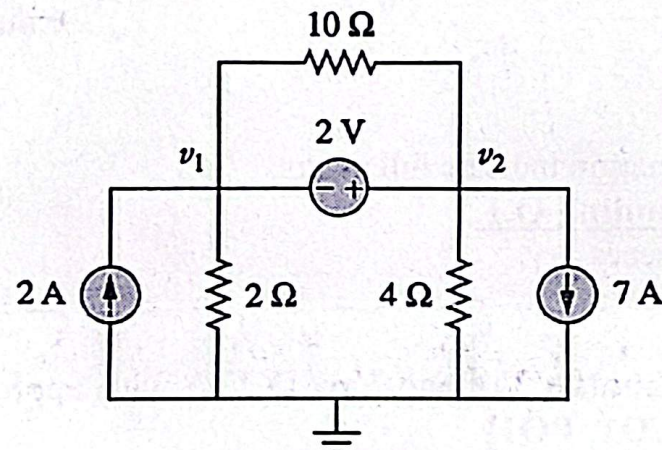


Fig. 03(b)

Course Code: HUM 1207

Course Title: Bangladesh Studies (History of Independence)

Full Marks: 20

Time: 01 Hour

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer two questions including Q.1.
3. Assume reasonable data if necessary.

- Q.1** The Partition of Bengal (1905) and the creation of a new province opened a new chapter in the history of this region. Whatever might have been the real motive of the colonial government behind the scheme, it divided the Hindus and the Muslims of Bengal.
- (a) Describe the background of Bengal partition in 1905 with historical examples. [CO1, PO6, C2] 3
- (b) Identify the key factors that led to the partition of Bengal, examining the political, economic, and religious influences to this historical event. [CO1, PO6, C2] 3
- (c) Demonstrate how the Partition of Bengal influenced Hindu-Muslim relations to create Lucknow pact. [CO1, PO6, C3] 4
- Q.2** (a) The Battle of Plassey became a decisive event in shaping colonial rule in India. Mention the main groups and leaders who took part in this historic battle. [CO1, PO7, C1] 2
- (b) Explain the underlying causes that led to the outbreak of the Battle of Plassey in 1757. [CO2, PO7, C2] 4
- (c) The aftermath of Plassey went far beyond the battlefield and transformed India's future. Sketch the major consequences of this battle and its role in expanding British control over Bengal and India. [CO2, PO7, C3] 4
- Q.3** (a) The Revolt of 1857 is considered the first large-scale resistance against British colonial rule in India. The uprising was not a sudden event but the result of years of political discontent and economic exploitation. State the political and economic causes of the revolt of 1857. [CO2, PO7, C1] 2
- (b) Identify the reasons behind the failures of the revolt of 1857 with historical examples. [CO2, PO7, C2] 4
- (c) The Revolt of 1857 was a turning point in Indian history. Its suppression brought about major administrative, political, and social changes in British India. Demonstrate about the consequences of the revolt of 1857. [CO2, PO7, C3] 4

Course Code: CE 1201

Full Marks: 20

Course Title: Engineering Mechanics

Time: 01 Hour

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer **two** questions **including Q.1.**
3. Assume reasonable data if necessary.

- Q.1 (a) For the Figure Q. 1 (a), sketch a free body of the members designated by the letter A and B. All plane surfaces and pins are smooth (frictionless). [CLO1, PLO2, C2] 5

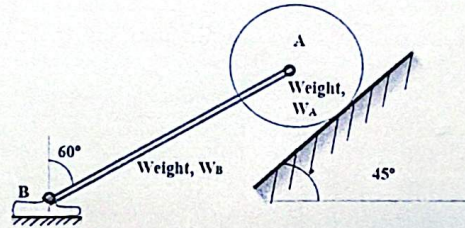


Figure Q. 1 (a)

- (b) A string ABCD attached to fixed points A and D has two equal weight of 1000 N attached to it at B and C. The weights rest with the portions AB and CD inclined at angles as shown in Figure Q. 1 (b). Find the tension in the portions AB, BC and CD of the string, if the inclination of the portion BC with the vertical is 120°. [CLO2, PLO2, C3] 5

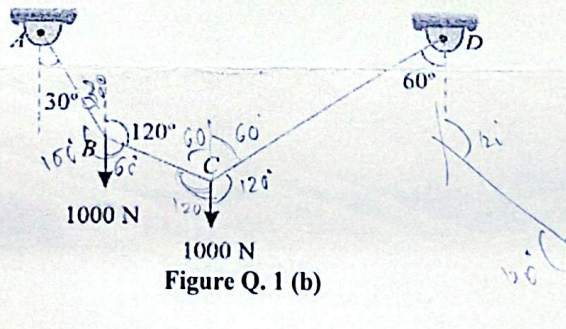


Figure Q. 1 (b)

- Q.2 (a) Calculate the support reactions for the beam shown in Figure Q. 2 (a). [CLO1, PLO2, C3] 5

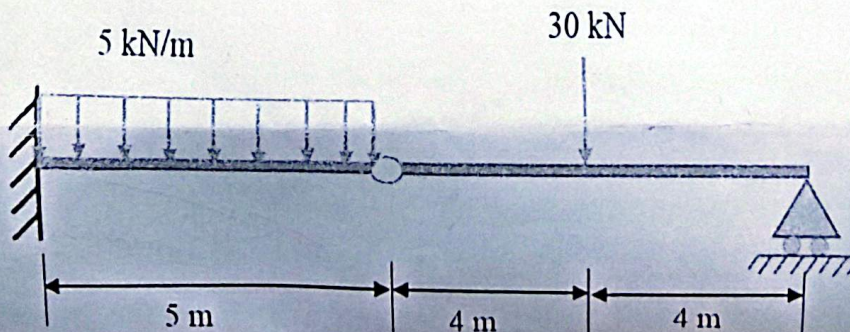


Figure Q. 2 (a)

- (b) If $F_B = 3 \text{ kN}$ and $\theta = 45^\circ$ in Figure Q. 2 (b), determine the magnitude of the resultant force of the two tugboats and its direction measured clockwise from positive x axis. [CLO1, PLO2, C3] 5

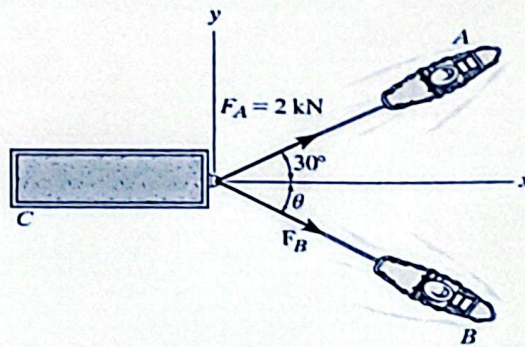


Figure Q. 2 (b)

Or

- Q.3 (a) Calculate the support reactions for the beam shown in Figure Q. 3 (a). [CLO1, PLO2, C3] 5

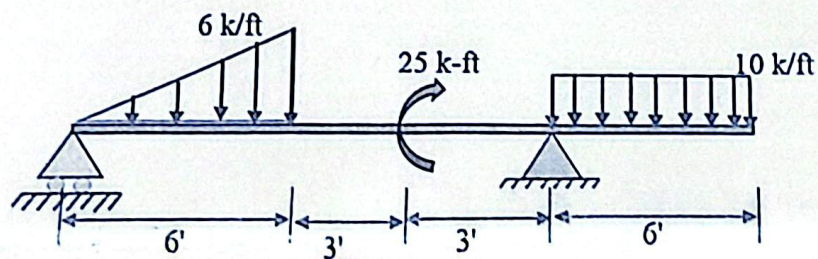


Figure Q. 3 (a)

- (b) Calculate the support reactions for the structure shown in Figure Q. 3 (b). [CLO1, PLO2, C3] 5

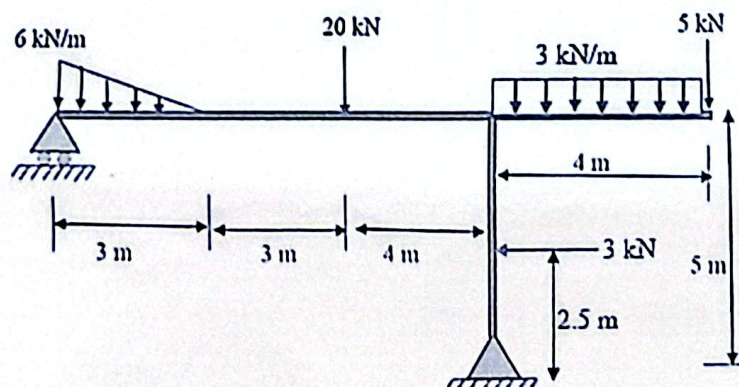


Figure Q. 3 (b)

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